



MedTech
Mediterranean
Institute of Technology

PRE-ENGINEERING PROGRAM

INT 101 FINAL REPORT

Development of a Full-Stack Web Service with AI-Powered WhatsApp Notifications for Automated News Distribution

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MEDIA**NET**

Tunis, 2025-2026

Approval

APPROVED BY

ACADEMIC SUPERVISOR

Name	Signature	Date
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COMPANY SUPERVISOR

Name	Signature	Date
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ACADEMIC EVALUATOR

Name	Signature	Date
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Declaration

I certify that I am the author of this project and that any assistance I received in its preparation is fully acknowledged and disclosed in this project. I have also cited any source from which I used data, ideas, or words, either quoted or paraphrased. Further, this report meets all the rules of quotation and referencing in use at MedTech, as well as adheres to the fraud policies listed in the MedTech honor code.

No portion of the work referred to in this study has been submitted in support of an application for another degree or qualification to this or any other university or institution of learning.

Student Name
Date

Signature

Work Term Release

I hereby state and verify by my signature that I have reviewed this report. I hereby affirm that the report contains

- ☐ no confidential data/information, and I authorize it to be released.
- ☐ confidential data/information, and I do not authorize it to be released.

COMPANY SUPERVISOR

Name	Signature	Date
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**Internship Agreement and Certificate Submission
Receipt
(To be placed here)**

Abstract

During my one-month internship at Medianet, a digital transformation agency in Tunisia with over 20 years of experience, I developed a complete web application that sends automated WhatsApp notifications for news updates and includes a smart chatbot. I worked on both frontend development using HTML, CSS, JavaScript, and React to create user-friendly interfaces, and backend development using Node.js, Express.js, and MySQL for data management. The main achievement was integrating Twilio's WhatsApp API for automatic notifications and creating a chatbot using Gemini API that answers user questions in multiple languages with the latest database information. Working with experienced professionals at Medianet taught me industry standards and project management while completing the project on time, giving me valuable experience in modern web development and API integration that will benefit my future career.

Keywords: Full-stack development, WhatsApp API, Chatbot, Gemini API, React.js, Node.js, Express.js, MySQL, Twilio, Medianet, Automated notifications.

Acknowledgements

I would like to express my sincere appreciation to the Medianet team for providing an excellent learning environment for interns. I am especially grateful to my supervisor, Mr. Hassen Benzarti, whose continuous guidance and support encouraged me to take initiative and develop solutions to the challenges we encountered during this internship.

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there is no equations.

1.EXECUTIVE SUMMARY

This report describes a web application developed during a one-month internship at Medianet that automatically sends news updates through WhatsApp and includes a chatbot for user questions. The project addressed the need for automated communication in digital marketing by creating a system that streamlines news distribution and enhances customer engagement. The solution helps businesses communicate with customers automatically while reducing manual work and improving response times through an admin interface, automated notifications, and a smart chatbot that answers queries in multiple languages.

Key recommendations for implementation include ensuring the system can handle many users, monitoring performance regularly, maintaining the database properly, and training staff on system administration. This project demonstrates how combining web development with AI technology can improve customer communication processes while maintaining efficiency for businesses in the digital transformation sector.

2.INTRODUCTION

Digital communication has transformed how businesses connect with customers, creating opportunities for automated engagement through platforms like WhatsApp and AI tools for customer support. This report presents a web application that automatically sends news updates through WhatsApp and includes a smart chatbot for user questions, developed during a one-month internship at Medianet, a digital transformation agency in Tunisia. The main purpose is to create a system that automatically distributes news content while providing users with an intelligent conversation interface. Key research questions include how web technologies work with messaging systems, what technical requirements are needed for smart chatbots, how businesses benefit from automated messaging with intelligent support, and what challenges occur when connecting multiple technologies. This project demonstrates the practical use of modern web development in creating solutions that improve customer engagement.

3.COMPANY CONTEXT

3.1. Description of the company

My internship took place at Medianet, a Tunisian company that was started in 1998 by four founders: Akrem Béji, Mohamed Mellouki, Iheb Béji, and Nidhal Battikh. When Medianet first began, it was a web agency that understood how important it was for Tunisian companies to communicate well with their customers online. The company started by focusing on web development, building attractive and working websites that fit each client's business needs.



Figure 1 : Medianet Logo

As digital marketing became more popular and important, Medianet expanded its services to include marketing. The company changed from just building websites to becoming a full web marketing agency that helps clients create their own marketing plans to promote their websites and reach more people online.

3.2. Mission and Objectives

3.3. Industry structure

3.4. Market Structure

4. INTERNSHIP DESCRIPTION

Describe your internship topic here.

4.1. Internship Context

4.2. General and specific objectives of the internship

4.3. Challenges and Obstacles

Describe any challenge/obstacles you faced during your project, and how you coped with them.

4.4. Assigned Tasks and Responsibilities

Summarize your project goals, activities, accomplishments, specific Tasks/assignments that you did alone or with others, and staff meetings that you take part in and your specific role/contribution in each meeting.

In addition, describe the key learning's you took away from your project. What company initiatives you observed, etc.)?

5.LITERATURE REVIEW

Relate your project topic to current literature or a theoretical framework.

6.METHODOLOGY

Choose strategies and methods in concordance with research- or investigation questions, then discuss, motivate and explain these choices: what did you use, a qualitative or quantitative research methods? Did you administer a questionnaire or interview people? Any field research conducted? How did you collect data? Did you utilize other libraries or archives?

7.RESULTS AND FINDINGS

8.RECOMMENDATIONS

This section should clearly and concisely state the issue facing the organization as substantiated by your analysis of the situation. Fundamental to determining the key issue is a statement of consequence regarding the potential impact for the firm if nothing is done (Explain the consequences if the key issues are not addressed by management). This should directly relate to the current situation of the company and the results/findings of your study.

9.CONCLUSIONS

The conclusion brings everything together. A stimulating and informative conclusion leaves the reader informed and well-satisfied.

REFERENCES

Generally, only references cited in the text are included in the references list; however, an occasional exception can be found to this rule. For example, supervisors may require evidence that students are familiar with a broader spectrum of literature than that immediately relevant to their study/report. In such instances, the reference list is called a bibliography.

All material must be referenced using the IEEE style as indicated in the MedTech Internship guide. All sources must be acknowledged in your reports each time you use a finding from someone's work. For more details: <https://ieeeauthorcenter.ieee.org/wp-content/uploads/IEEE-Reference-Guide.pdf>

APPENDIX

The following materials are appropriate for an appendix: company specific data/information, market data, material that is additional to the information supplied in the main write up.